

Gene expression

scDetect: a rank-based ensemble learning algorithm for cell type identification of single-cell RNA sequencing in cancer

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Abstract

Motivation: Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of different cell types in many tissues and tumor samples. Cell type identification is essential for single-cell RNA profiling, currently transforming the life sciences. Often, this is achieved by searching for combinations of genes that have previously been implicated as being cell-type specific, an approach that is not quantitative and does not explicitly take advantage of other scRNA-seq studies. Batch effects and different data platforms greatly decrease the predictive performance in inter-laboratory and different data type validation.

Results: Here, we present a new ensemble learning method named as ‘scDetect’ that combines gene expression rank-based analysis and a majority vote ensemble machine-learning probability-based prediction method capable of highly accurate classification of cells based on scRNA-seq data by different sequencing platforms. Because of tumor heterogeneity, in order to accurately predict tumor cells in the single-cell RNA-seq data, we have also incorporated cell copy number variation consensus clustering and epithelial score in the classification. We applied scDetect to scRNA-seq data from pancreatic tissue, mononuclear cells and tumor biopsies cells and show that scDetect classifies individual cells with high accuracy and better than other publicly available tools.

Availability and implementation: scDetect is an open source software. Source code and test data is freely available from Github (<https://github.com/IVDgenomicslab/scDetect/>) and Zenodo (<https://zenodo.org/record/4764132#.YKCOIrH5AYN>). The examples and tutorial page is at <https://ivdgenomicslab.github.io/scDetect-Introduction/>. And scDetect will be available from Bioconductor.

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Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

Single-cell RNA-sequencing (scRNA-seq) has emerged as a revolutionary technique, which can be used to identify the unique transcriptomic profile of each cell. Using this information, we are now able to address questions that previously could not be answered, including the identification of new cell types (Grün *et al.*, 2015; Segerstolpe *et al.*, 2016; Treutlein *et al.*, 2014; Villani *et al.*, 2017), resolving the cellular dynamics of developmental processes (Li *et al.*, 2017; Petropoulos *et al.*, 2016; Rizvi *et al.*, 2017) and determining

gene regulatory mechanisms that vary between cell subtypes (Treutlein *et al.*, 2016). Cell type identification and discovery of subtypes has emerged as one of the most important early applications of scRNA-seq (Aviv *et al.*, 2017).

Prior to the development of scRNA-seq, the traditional methods of cell classification relied on microscopy, histology and pathological criteria (Arendt *et al.*, 2016). In the field of immunology, cell surface markers have been widely used to distinguish cell subtypes for a wide range of purposes (Papalexi and Satija, 2018). While this approach is desirable in practical terms for cell isolation [e.g.

fluorescence-activated cell sorting (FACS)], these markers may not reflect the overall heterogeneity at a transcriptomic and phenotypic level from mixed cell populations (Grün and van Oudenaarden, 2015; Trapnell, 2015). Unsupervised and supervised clustering approaches have been used to determine groups of cells based on similar transcriptional signatures within a sample (Kiselev et al., 2017; Macosko et al., 2015), and frequently, cells within a cluster are collectively labeled based on the average expression levels of canonical markers (Stegle et al., 2015).

Recently, a number of cell type identification algorithms have appeared in the literature to address these problems. Automated methods, such as scPred (Alquicira-Hernandez et al., 2019), CHETHA (de Kanter et al., 2019) and SingleR (Aran et al., 2019), base their cell type identification capacity on comparisons with annotated reference data using automatically chosen genes that optimally discriminate between cell types. A good cell type identification method should be both sensitive and selective with the capacity to correctly identify as many cells as possible, while not classifying cells when there is insufficient evidence. If the cell being identified is of a type not represented in the reference set, such misclassification can easily occur. This is a concern when studying malignant cells, which are often too heterogeneous to include in the reference data. To avoid over-classification, methods such as scPred (Alquicira-Hernandez et al., 2019) and CHETHA (de Kanter et al., 2019) leave cells unclassified if they are too dissimilar to any reference data.

Research into tumor composition presents an even more challenging setting, since the RNA expression profile of malignant cells is often different from any known cell type, and can also be unique to the patient and biopsy (Puram et al., 2017; Tirosh et al., 2016a,b). Malignant cells can sometimes be identified in scRNA-seq data (Patel et al., 2014) through copy number variations and epithelial origin. In addition, a first sign of the malignancy of cells in the sample is their imperviousness to classification, simply because their expression profiles do not resemble that of any known, healthy cell type.

Here, we present scDetect, a method to quantitatively classify scRNA-seq data based on comparison to a reference dataset. Because of the batch effect and differences in the nature of the data acquired by RNA-seq- and microarray-based measurements, the predictive performances of biomarkers based on gene expression profiles often decrease greatly in inter-laboratory validation. To tackle this problem and make query and reference data compatible across platforms, we use a transformation based on comparing the expression of pairs of genes within each cell, a method inspired by the top-scoring pair classifier (Geman et al., 2004). Here, we evaluate the performance of scDetect, compare it to the intermediate quantitative outputs of three existing classification methods: SingleR, CHETHA and scPred. To more accurately predict the malignant cells in the single-cell RNA-seq data, we further incorporated cell copy number variation information and epithelial origin information in the classification method.

2 Materials and methods

2.1 The scDetect algorithm

TSP (Top Scoring Pairs) classifier was introduced by Geman et al. (2004) (Geman et al., 2004) for the classification of gene expression data based entirely on relative gene expression values, specifically pairwise comparisons between two gene expression levels. In essence, the program exploits discriminating information contained in the R matrix by focusing on ‘marker gene pairs’ (i, j), for which there is a significant difference in the probability of the event across the N samples from class C_1 to C_2 . The quantities of interest are $p_{ij}(C_m) = \text{Prob}(R_i < R_j | Y = C_m)$, $m = \{1, 2\}$. These probabilities are estimated by the relative frequencies of occurrences of $R_i < R_j$ within profiles and over samples. Letting Δ_{ij} denote the ‘score’ of the gene pair (i, j), where $\Delta_{ij} = |p_{ij}(C_1) - p_{ij}(C_2)|$, the method computes the score Δ_{ij} for every pair of genes $i, j \in \{1, \dots, P\}$, $i \neq j$. Pairs of genes with high scores are viewed as most informative for classification. For each top-scoring gene pair (i, j), the method computes the ‘average rank difference’ γ_{ij} in class C_m , defined as

$$\gamma_{ij}(C_m) = \frac{\sum_{n \in C_m} (R_{i,n} - R_{j,n})}{|C_m|}, \quad m = \{1, 2\}$$

Based on the original TSP algorithm, k -TSP, an ensemble method that uses K pairs of genes for classifying gene expression data (Tan et al., 2005). When $k = 1$, this algorithm, referred to simply as TSP necessarily selects a unique pair of genes. More generally, both TSP and k -TSP may be seen as special cases of a new classification methodology based on the concept of ‘relative expression reversals’.

However, both TSP and k -TSP are designed for binary classification problems. Therefore, in this study, we developed a weighted ensemble k -TSP algorithm for multi-class classification in single-cell RNA-seq data cell type detection (scDetect). Three steps were included in the scDetect algorithm for identifying the cell types of each cell (Fig. 1A).

Step 1: Feature genes identification

We first selected the top 2000 most informative genes measured by median absolute deviation (MAD) to generate a dataset including the variable genes. We identified the feature genes for each cell type based on a ‘One-vs-Others approach’. Given multiple cell types $T = \{T_1, T_2, \dots, T_x\}$, the One-vs-Others approach decomposes the original problem into a set of M two-class problems. For each cell type $x = 1, \dots, X$, we trained the classifier and identified the top score gene pairs as the feature genes based on k -TSP method for distinguishing between the individual cell type T_x and the composite cell types consisting of all other classes (Fig. 1A). To select the number of pairs, we measured the accuracy on the training set by calculating the area under the receiver operating characteristic curve (AUC) for each possible number of pairs K .

Step 2: Cell type prediction score calculation

A weighted majority vote k -TSP algorithm was developed for extending binary to multi-class cell type classification. A binary k -TSP cell type classifier is constructed for each distinct pair of classes cell type $T_x, T_y \in T$, $T_x \neq T_y$, using only the training samples for those cell types. Consequently, this approach generates $X(X-1)/2$ binary cell type classifiers (Fig. 1A). We used all of the identified genes in Step 1, after removing the redundant ones, to train each classifier. In this scheme, the cell type classifiers were combined by weighted voting which is based on the score from the prediction results of each classifier (Fig. 1A). Finally, for each cell, we obtained a prediction score of each cell type.

Step 3: Confidence score generation and cell type classification

To further increase the precision of the method, we calculate the confidence score of the prediction score of each cell based on the permutation analysis. We randomly selected the same number of genes among all genes 1000 times and calculated the prediction score using the method in Step 2 to generate a null distribution of the prediction score. By calculating the proportion of the score which is equivalent to or higher than the actual score, we obtain a P value for the likelihood of cell-type prediction score reliability. In other words, if the observed prediction score is in the top 5%, the result is assigned a confidence score with a P value = 0.05. The final prediction results have two states. If the confidence score of the cell type prediction score is lower than confidence threshold (P value > 0.05), the result is placed in the ‘Unknown’ category. If the confidence score is higher than confidence threshold (P value < 0.05), the final result is the cell type that has the highest score among all the cell types.

2.2 scDetect algorithm for cancer scRNA-seq data

Although the scDetect method had good performance in cell type classifications in single-cell RNA-seq data across different datasets, it is still a challenge to accurately predict cancer cells in single-cell RNA-seq data in tumor samples. Because of tumor heterogeneity in the cancer samples, cancer cells in different samples had distinct expression patterns, which decreased the ability of scDetect to predict the cell types. To improve the prediction precision of cancer cells in single-cell RNA-seq data, we incorporated the cell type classification results of scDetect, copy number variation (CNV) consensus clustering and epithelial score into the classification framework (Fig. 1B).

There are three steps involved in the scDetect-Cancer (Fig. 1B). Step1: scDetect was first used to classify the immune T cells as the normal cell reference for the CNV and epithelial score analysis in next step. Step2: We performed CNV consensus clustering and calculated the epithelial gene set enrichment score of each cell based on the normal reference cells to predict the tumor cells. The cells which had both copy number variation and high epithelial score were determined as the tumor cells. Step3: We used scDetect to predict the cell type of other cells. If the cell had both unknown cell type and copy number variation, we classified the cell as the tumor cell and the other cells will be classified as immune and stromal cell types.

Copy number variation (CNV) consensus clustering

Step 1: Copy number variations (CNVs) were estimated by sorting the analyzed genes by their chromosomal location and applying a moving average to the relative expression values, with a sliding window of 100 genes within each chromosome, as previously described (Patel *et al.*, 2014; Tirosch *et al.*, 2016a,b). To avoid considerable impact of any particular gene on the moving average, we limited the relative expression values to $[-3, 3]$ by replacing all values above 3 by a ceiling of 3, and replacing values below -3 by a floor of -3. InferCNV was used to analyze tumor single-cell RNA-Seq data to identify evidence for large-scale chromosomal copy number variations, such as gains or deletions of entire chromosomes or large segments of chromosomes. This was done by analyzing expression intensity of genes across positions of the genome in comparison to the average or a set of reference 'normal' cells. Step 2: Consensus Cluster Plus R-package (Wilkerson and Hayes, 2010) was used to identify clusters in the data using 1000 iterations, 80% sample re-sampling from 2 to 10 clusters using hierarchical clustering with average inner Linkage and final Linkage and Pearson correlation as the similarity metric. Step 3: To automatically determine which cluster is the tumor cell cluster among the k clusters found, we used the Shannon Entropy (SE) to quantify the specificity of CNV across all the regions in each cell. We assign:

$$P(x_i) = \frac{S_i}{S_n}$$

where S_i represents the relative amplification/deletion level of a region on the genome in one cell, and S_n is the total relative amplification/deletion level in the selected cell. Thus, $P(x_i)$ represents the percentage of CNV level in the i th region among the CNV levels of the genome in the cell. Building upon this premise, we propose:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

where $H(X)$ represents Shannon Entropy. The greater the uncertainty of the variable is, the larger the Shannon Entropy generated. Therefore, when the SE is larger, the observed CNV result from InferCNV is more random in the cell (normal cells). In contrast, the closer SE is to 0, the higher the CNV specificity of the regions in the genome (tumor cells). We take the cluster with the highest SE as the normal cell cluster. And the other clusters will be regarded as tumor cell clusters.

Epithelial score calculation

We first identified a set of potential epithelial genes based on the results of gene expression differential analysis between tumor cells and normal cells in melanoma and head neck cancer single-cell data. We excluded potential genes that had low expression or were not co-regulated with the other markers across all single cells (Pearson $R < 0.4$ with the average of all other markers). We found a total of 1579 epithelial related genes which are differentially expressed between tumor and normal cells in melanoma and head-neck data. To identify cells with active epithelial gene sets in single-cell RNA-seq data, AUCell was used to perform the analysis to quantify the epithelial score (Aibar *et al.*, 2017). We calculated the enrichment of the epithelial gene sets as an area under the recovery curve across the ranking of all genes in a particular cell, whereby genes are ranked by their expression value. Therefore, this method is independent of the gene expression units and normalization procedure. The distribution

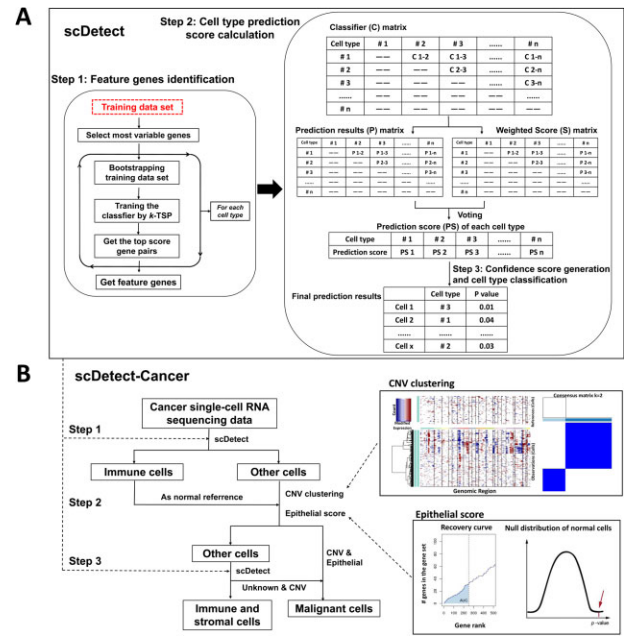


Fig. 1. Summary of the scDetect algorithm. (A) scDetect algorithm for scRNA-seq data in normal tissues. Three steps are included in the scDetect algorithm for identifying the cell types of each cell. Step 1: Feature gene identification. We first selected the most informative genes measured by median absolute deviation (MAD) to generate a dataset including the variable genes. We identified the feature genes for each cell type based on a 'One-vs-Others approach'. Step 2: Cell type prediction score calculation. A weighted ensemble k-TSP algorithm was developed for extending binary to multi-class cell type classification. A binary cell type classifier is constructed for each distinct pair of classes cell type. Consequently, this approach generates $X(X-1)/2$ binary cell type classifiers. The cell type classifiers were combined by weighted voting which is based on the score from the prediction results of each classifier. Finally, for each cell, we obtained a prediction score of each cell type. Step 3: Confidence score generation and cell type classification. To further increase the precision of the method, we calculate the confidence score of the prediction score of each cell based on the permutation analysis. We randomly selected the same number of genes among all genes 1000 times and calculated the prediction score using the method in Step 2 to generate a null distribution of the prediction score. The final prediction results have two states. If the confidence score of the cell type prediction score is lower than confidence threshold (P value > 0.05), the result is placed in the 'Unknown' category. If the confidence score is higher than confidence threshold (P value < 0.05), the final result is the cell type that has the highest score among all the cell types. (B) scDetect algorithm for scRNA-seq data in cancer samples. Three steps involved in the scDetect-Cancer (B). Step1: scDetect was first used to classify the immune T cells as the normal cell reference for the CNV and epithelial score analysis in next step. Step2: We performed CNV consensus clustering and calculated the epithelial gene set enrichment score of each cell based on the normal reference cells to predict the tumor cells. The cells which had both copy number variation and high epithelial score were determined as the tumor cells. Step3: We used scDetect to predict the cell type of other cells. If the cell had both unknown cell type and copy number variation, we classified the cell as the tumor cell and the other cells will be classified as immune and stromal cell types

of epithelial scores across all the cells allows analyzing the relative expression of the epithelial gene sets. Since the scoring method is ranking-based, this method is independent of the gene expression units and normalization procedure. After the epithelial score of all cells were calculated, we used the normal immune reference cells as the null distribution to calculate the differences (P value) between each cell and the normal distribution.

2.3 Datasets

A total of 20 scRNA-seq datasets (gene-cell raw count matrix) were used to evaluate and benchmark scDetect and other classification methods (Supplementary Table S1). From this six datasets were used for intra-dataset evaluation using a cross-validation scheme and all datasets were used for inter-dataset evaluation. Datasets vary across tissue (pancreas, PBMC and cancer) and the sequencing protocol used for analysis. All four pancreatic datasets were obtained from

the GEO database (Baron Human: GSE84133, Muraro: GSE85241, Segerstolpe: EMTAB-5061, Xin: GSE81608) (Baron *et al.*, 2016; Muraro *et al.*, 2016; Segerstolpe *et al.*, 2016; Xin *et al.*, 2016). All 13 peripheral blood mononuclear cells (PBMC) datasets were downloaded from the Broad Institute Single Cell portal (Ding *et al.*, 2019). The cancer single-cell RNA-seq data were downloaded from the GEO database (Puram *et al.*, 2017; Tirosh *et al.*, 2016a,b). The cell population annotations for all datasets were provided with the data. These annotations were used as a ‘ground truth’ during the evaluation of the cell population predictions obtained from the classification methods.

2.4 Evaluation of scDetect algorithm based on intra/inter-dataset classification

To evaluate the performance of the scDetect and compare its performance relative to previous methods, we performed the intra/inter-dataset classification analysis on scDetect and three other previously reported cell type classification methods, namely ‘SingleR’ (Aran *et al.*, 2019), ‘scPred’ (Alquicira-Hernandez *et al.*, 2019) and ‘CHETHA’ (de Kanter *et al.*, 2019), in multiple datasets. During the benchmark, all methods were run using their default settings. Where this option was not available, we used the settings provided in the accompanying examples or vignettes. As input, we provided each method with the raw count data according to the method documentation.

Data preprocessing

Based on the manual annotation provided in the datasets, we started by filtering out cells that were labeled as doublets, debris or unlabeled cells. Next, we filtered genes with zero counts across all cells. The number of cells and genes in Supplementary Table S1 represent the size of each dataset after this stage of preprocessing. Moreover, before applying cross-validation to evaluate each classifier, we excluded cell populations with less than 10 cells across the entire dataset. Supplementary Table S1 summarizes the number of cell populations before and after this filtration step for each dataset.

Intra-dataset classification

For the supervised classifiers, we evaluated the performance by applying a 5-fold cross-validation across each dataset after filtering genes, cells and small cell populations. The folds were divided in a stratified manner in order to keep equal proportions of each cell population in each fold. The training and testing folds were exactly the same for all classifiers.

Inter-dataset classification

We selected the four major endocrine pancreatic cell types (alpha, beta, delta and gamma) across all four human pancreatic datasets: Baron Human, Muraro, Segerstolpe and Xin. We applied leave-one dataset-out cross-validation where we train on three datasets and test on the fourth, left-out dataset. Because the pancreas datasets were not included the common 10× Chromium data, we further tested the classification performance on the PBMC datasets, which included two 10× Chromium datasets and other 5 datasets from different platforms (CEL-Seq2, Drop-seq, inDrops, Seq-Well, Smart-seq2). We further evaluate the performance of the scDetect and other three methods in non-malignant cells identified in a cancer microenvironment across melanoma, head-neck cancer and colon cancer samples.

Performance evaluation metrics

For each cell, the predicted cell type was compared with the reference cell type annotation. A true-positive result was indicated when the predicted cell type matched the reference cell type. When the predicted cell type and reference cell type did not match, the cell was considered a false positive. For each cell type, recall was defined as the ratio of true-positive results to the total positive samples analyzed, while precision was defined as the ratio (1 - false positive)/(total tested total positive). The performance of the methods on the datasets is evaluated using three different metrics: (i) For each cell population in the dataset, the prediction accuracy is used as a measure for the performance on the dataset. The accuracy was defined as

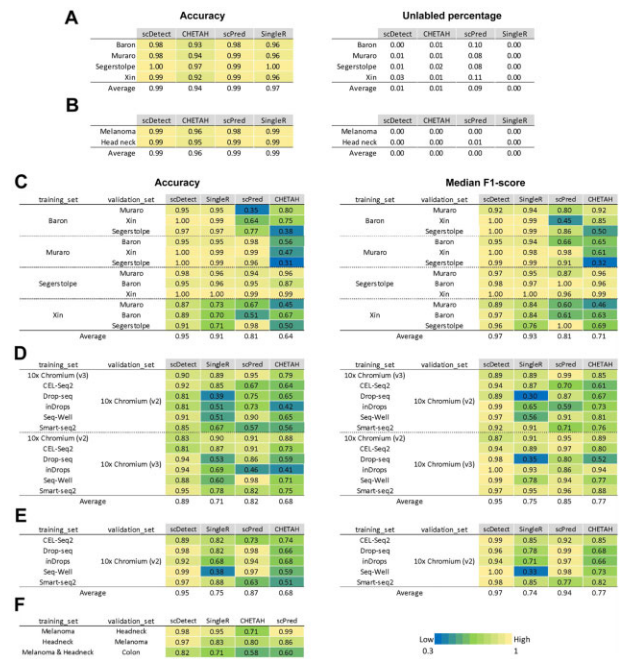


Fig. 2. Performance comparison of scDetect and other methods for cell identification using different scRNA-seq datasets. Heatmap of the intra-dataset evaluation accuracy and percentage of unlabeled cells across all cell populations per classifier (rows) per dataset (columns) in (A) pancreas datasets, (B) cancer microenvironment datasets. Heatmap of the inter-dataset evaluation accuracy and median F1-score across all cell populations per classifier per dataset in (C) pancreas datasets, (D) PBMC1 and (E) PBMC2 datasets. F, cancer microenvironment datasets

(true positive + true negative)/(total tested samples). (ii) For each cell population in the dataset, the F1-score is reported. The F1-score was defined as $2(\text{precision} \times \text{recall})/(\text{precision} + \text{recall})$. The median of these F1-scores is used as another measure for the performance on the dataset. F-score is a measure of a test's accuracy. It considers both the precision p and the recall r of the test to compute the score. (iii) Some of the methods do not label all the cells. These unassigned cells are not considered in the accuracy or F1-score calculation. The percentage of unlabeled cells is also used to evaluate the performance.

3 Results

scDetect is a generalized method for classifying a single cell based on its transcriptional profile data. The method uses a combination of gene expression rank-based classification methodology and a weighted majority vote machine learning approach to estimate cell type of each cell. In doing so, it is able to minimize (or eliminate) the batch effects and different data platform problems which greatly decrease the predictive performances of biomarkers based on gene expression profiles in inter-laboratory and different data type validations. First, we identified the feature genes based on training scRNA datasets. Second, a weighted ensemble learning method was developed for extending binary to multi-class cell type classification. A binary cell type classifier is constructed for each distinct pair of classes. The application of the prediction model to single-cell data obtained from independent sample, with each cell then assigned a cell type prediction score of belonging to a given cell subtype or state. Third, scDetect incorporates a rejection option to avoid assigning cells to a class when the conditional class probability is lower than a given value. We calculate the P value of the prediction score of each cell based on the permutation data null distribution. This means that the value should be higher than a probability threshold for a cell to be assigned to any respective class. When the maximum probability across all classes is lower than the threshold, then a cell is labeled as ‘Unknown’. This criterion avoids misclassifying cells

when some cell types are not considered in the prediction model but are present in the test data. For all of the analyses performed in this paper, we used a threshold of P value = 0.05. Here, we present the results of the application of scDetect under four distinct scenarios.

3.1 scDetect can accurately predict cell types in intra-dataset

We evaluated the performance of scDetect and benchmarked the performance of three other classifiers (i.e. 'SingleR', 'scPred' and 'CHETHA') across six datasets used for intra-dataset fivefold cross validation evaluation (Fig. 2). The datasets used in this study include four human pancreatic datasets (Baron Human, Muraro, Segerstolpe and Xin), which vary in the sequencing protocol used to generate the dataset (Supplementary Table S1). To further investigate the performance of scDetect in cancer microenvironment, we also included the melanoma and head-neck non-malignant cells in the validation datasets. Using the scDetect algorithm to classify each cell of the 4 pancreas datasets, we achieved an accuracy of 0.98 in the Baron dataset, 0.98 in the Muraro dataset, 1.00 in the Segerstolpe dataset and 0.99 in the Xin dataset, for an average prediction accuracy of cell type of 0.99. Compared to the other three methods, we found all classifiers perform well in the pancreas dataset, with the average prediction accuracy of CHETHA, scPred and SingleR being 0.94, 0.99 and 0.97, respectively.

The scDetect, CHETHA and scPred classifiers incorporate a rejection option to identify cell populations in the test set that were not seen during training. These populations cannot be predicted correctly and, therefore, should remain unassigned. We then evaluated the unlabeled cell percentage of the three classifiers in the four pancreas datasets. We found that the average unlabeled cell percentage was 0.01, 0.01 and 0.09 in scDetect, CHETHA and scPred, respectively. These results suggested that scDetect (0.99) has a higher prediction accuracy than CHETHA (0.94) and SingleR (0.97) in pancreas datasets. Compared with scPred, scDetect has comparable prediction accuracy and could accurately predict more cells (Unlabeled percentage: 0.01 in scDetect, 0.09 in scPred). To further evaluate the performance of the scDetect and other three methods in cancer microenvironments, we performed the intra-dataset fivefold cross validation in non-malignant cells of melanoma and head-neck cancer samples. The results showed that all four methods have high prediction accuracy (>0.95) and low unlabeled sample numbers in melanoma and head-neck non-malignant cells.

To evaluate the time complexity of scDetect and other algorithms, we performed fivefold cross validation of each method and calculated the time cost of each method in each fold. We selected two human pancreas single-cell data which have different cell numbers and cell population numbers to perform this analysis (Xin dataset: No. of cells: 1449, No. of cell populations: 4; Baron dataset: No. of cells: 8569, No. of cell populations: 14). In Xin dataset, the average time cost of each fold in each method is: CHETHA: 32.9 s, scPred: 57.7 s, SingleR: 20.3 s; and scDetect: 8.0 s (Supplementary Fig. S1). The results showed that for the Xin dataset, scDetect is the fastest method among the four softwares. With the Baron dataset, the average time cost of each fold in each method is: CHETHA: 69.8 s, scPred: 334.7 s, SingleR: 262.2 s; and scDetect: 145.7 s (Supplementary Fig. S1). The results showed that with the Baron dataset, scDetect is the second fastest method among four software compared with scDetect being slower than CHETHA, but faster than scPred and SingleR. The results with the Xin and Baron datasets suggest that the scDetect is the fastest approach when the cell number/cell population number is small. When cell number/cell population number increase, the scDetect becomes slower than the CHETHA method, but is still faster than scPred and SingleR.

3.2 scDetect can accurately predict cell types across different platforms

While it is important to evaluate the classification performance within a dataset (intra-dataset), a more important real-world evaluation is its capacity for cross-datasets and platforms (i.e. inter-dataset) classification. Therefore, we used 19 datasets (Supplementary Table

S1) to test the classifiers' ability to predict cell identities in a dataset that was not used for training (Fig. 2). First, we tested the performance of scDetect and the other three methods across the 4 pancreas datasets. All 4 pancreas dataset were generated using a different sequencing protocol (i.e. Baron: inDrop; Muraro: CEL-Seq2; Segerstolpe: SMART-Seq2; and Xin: SMARTer). We evaluated the classification performance when training on one pancreas dataset and testing on the other. Using the scDetect algorithm to classify each cell across the 4 pancreas datasets, we achieved an average accuracy in prediction of cell type of 0.95 and the median F1 score of 0.97. Compared with SingleR (Accuracy: 0.91, median F1 score: 0.93), scPred (Accuracy: 0.81, median F1 score: 0.81) and CHETHA (Accuracy: 0.64, median F1 score: 0.71), scDetect had better performance in analyzing these 4 pancreas inter-datasets. The results also showed that the classifiers had different performance in different training and validation datasets. When using the the Segerstolpe dataset as the training set, all four methods showed good performance ability to predict cell types in the other datasets. When using the Muraro dataset as the training set, the performance of the CHETHA method was not as good as the other three methods. When using the Baron dataset as the training set, only two methods, scDetect and SingleR, had good performance. When we used the Xin dataset as the training set, the performance of all four methods was not very good.

Because the pancreas dataset were not included in the common 10× Chromium data, we further tested the classification performance on the PBMC datasets, which included two 10× Chromium datasets and 5 other datasets generated from different platforms (i.e. CEL-Seq2, Drop-seq, inDrops, Seq-Well and Smart-seq2). The PBMC dataset consists of 2 samples (PBMC1 and PBMC2), sequenced using 7 different protocols (Supplementary Table S1) with the exception that 10Xv3 was not applied to the PBMC2 sample. To evaluate the performance of the four classification methods, we used the 10Xv2, 10Xv3 as the validation set and other dataset as training set. Using the scDetect algorithm to classify each cell across the PBMC1 datasets, we achieved an average accuracy in prediction of cell type of 0.89, and the median F1 score is 0.95. Compared with SingleR (Accuracy: 0.71, median F1 score: 0.75), scPred (Accuracy: 0.82, median F1 score: 0.85) and CHETHA (Accuracy: 0.68, median F1 score: 0.77), scDetect had better performance in PBMC1 inter-data experiment. In the PBMC2 datasets, the prediction average accuracy is 0.95 and the median F1 score is 0.97. Compared with SingleR (Accuracy: 0.75, median F1 score: 0.74), scPred (Accuracy: 0.87, median F1 score: 0.94) and CHETHA (Accuracy: 0.68, median F1 score: 0.77), the results suggested that scDetect also had better performance in PBMC2 inter-data experiment, which consistent with results in PBMC1 datasets. The results also showed that the classifiers had different performance abilities in different training and validation datasets. When using the 10Xv3 dataset as the training set to predict the cell types in 10Xv2 in PBMC1 datasets, the scPred classifier had higher accuracy (0.95) than other methods. When we used other datasets (CEL-Seq2, Drop-seq, inDrops, Seq-Well, Smart-seq2) as the training sets, the scDetect had the better performance than other methods. And the results are also consistent in PBMC2 datasets.

We further evaluated the performance of scDetect and other three methods using cancer microenvironment datasets that include melanoma, head-neck cancers and non-malignant cells from colon cancer samples. When using the melanoma as training set and head-neck data as the validation dataset, the results showed that scDetect (0.98) and scPred (0.99) had higher prediction accuracy than either SingleR (0.95) or CHETHA (0.71). When using head-neck data as the training set, the results showed that the scDetect method (0.97) had better performance than any of the other three methods (scPred: 0.83, SingleR: 0.80, CHETHA: 0.86). When we combined the melanoma and head-neck non-malignant cells together as the training dataset to predict the cell types in non-malignant colon cancer sample cells, scDetect archived the highest accuracy (0.82) among the four methods.

3.3 scDetect-cancer can accurately predict cell types in cancer scRNA data

Because the cancer samples contain tumor heterogeneity, cancer cells in different samples had distinct expression patterns, which decreased the performance of classification method to predict the cell types. Although the scDetect method had good performance in cell type classifications in single-cell RNA-seq data across different datasets, it is still a challenge to accurately predict cancer cells in cancer single-cell RNA-seq data. In order to help improve prediction accuracy in the single-cell RNA-seq data, we incorporated cell copy number information and epithelial level information (two types of information generated by the single-cell RNA-seq data) in the classification framework (scDetect-cancer) for cell type classification in cancer scRNA data. Three steps were included in the scDetect-cancer classification framework. First, we used scDetect to classify the immune T cells as the normal cell reference for the CNV and epithelial score analysis in next step. Second, we calculated the copy number variation score and epithelial genes enrichment score of all cells based on the normal reference cells. In this step, we performed consensus clustering analysis in CNV data and permutation analysis in epithelial score data to accurately predict the tumor cells. The cells which had both copy number variation and significantly higher epithelial score were classified as the malignant cells. Third, all of the other cells were classified by scDetect. If the cell had both unknown cell type and copy number variation, we classified the cell as the malignant cell and the other cells were classified as immune and stromal cell types.

To incorporate the copy number variation information of each cell in the classification analysis, we defined a set of immune cells as the CNV calculation normal reference. To investigate which cell types of the immune cells could be more precisely predicted by scDetect in the tumor samples, we performed a classification experiment in the melanoma and head neck cancer samples using the immune cells as the training data. In the melanoma data, the prediction accuracy of T cells (0.99) is significantly higher than that of other immune cell types (e.g. Macrophage: 0.91, NK: 0.81, B cell: 0.86). In the head-neck cancer dataset, the prediction accuracy of T cells (0.99) is also higher than that of other immune cell types (Macrophage: 0.94, Bcell: 0.92). These results suggested that scDetect could accurately predict immune T cells in single-cell tumor samples based on using the immune cells as the training data. Therefore, the immune T cells were used as the normal reference in the next step of analysis. After defining the normal immune T cells as the reference cells, we used single-cell RNA-Seq data to identify evidence for large-scale chromosomal copy number variations in each cell (Fig. 3A, F). This was done by analyzing expression intensity of genes across positions of the genome in comparison to the average or a set of reference 'normal' cells. Based on the results, we found the regions of the genome which are over-abundant or less-abundant as compared to normal cells.

To further determine the best cluster number k , we performed consensus clustering analysis of the CNV data. We applied different k values to cluster the single cells based on the CNV data, and calculated the cluster-consensus value of each cluster in each k . The cluster-consensus value is the mean of all pairwise consensus values between members of a cluster. High values indicate a cluster has high stability and low values indicate a cluster has low stability. We classified the samples into k clusters when the average cluster-consensus score of the k is maxim. In the melanoma data, we found that when the $k=2$, the average cluster-consensus score of the two clusters is 1, which higher than the average cluster-consensus score in other k (Fig. 3B, C). In the head neck cancer data, when the $k=4$, the average cluster-consensus score is higher than other k value (Fig. 3G, H). Therefore, the head neck cancer single cells fall into four clusters based on the CNV information. To determine which cluster is the tumor cell cluster among the k cluster we found, we used the Shannon Entropy to quantify the specificity of CNV across all the regions in each cell (CNV SE-score). We calculated the average CNV SE-score of each cluster based on the score of each cell in the cluster. In melanoma data, the results showed that the two cell clusters had distinct CNV SE-score distributions. The CNV SE-score

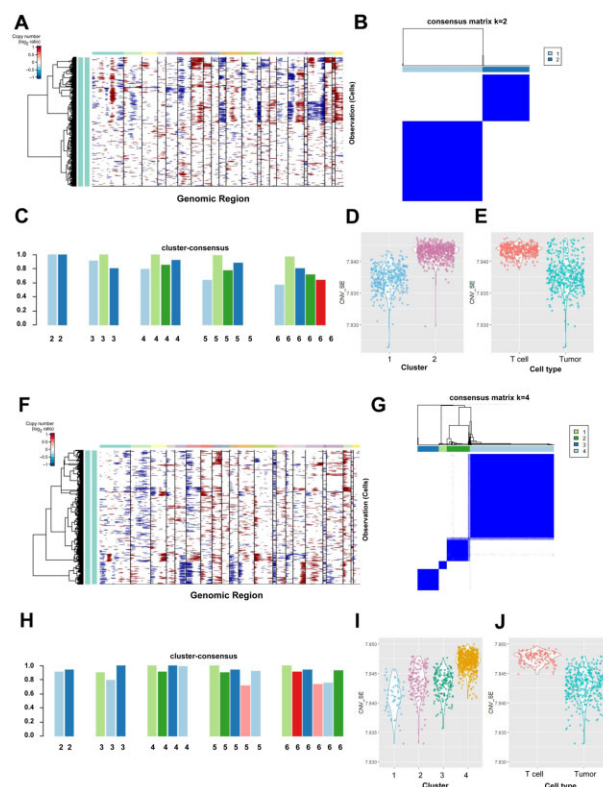


Fig. 3. Copy number variation consensus clustering for tumor cell identification. (A–E) are melanoma datasets and (F–J) are head-neck datasets. (A, F) Heatmap showing large-scale CNVs for individual cells (rows). Red: amplifications; blue: deletions. (B, G) The heatmap of cluster consensus matrix for $k=2$ for the melanoma datasets and $k=4$ for the head-neck datasets. The matrix has items as both rows and columns and where consensus values range from 0 (never clustered together) to 1 (always clustered together) marked by white to dark blue. (C, H) Cluster-consensus value of clusters at each k . This is the mean of all pairwise consensus values between a cluster's members. Clusters are indicated following the same color scheme as used in the cluster matrices and tracking plots. The bars are grouped by k which is marked on the horizontal axis. High values indicate a cluster has high stability and low values indicate a cluster has low stability. (D, I) Violin plot shows distributions of CNV SE-score (Shannon Entropy) for cells categorized in each cluster. (E, J) Violin plot shows distributions of CNV SE-score for T cells and tumor cells

is significantly higher in cluster2 than cluster1 (Fig. 3D), indicating that cluster1 might be the tumor cell cluster and cluster2 might be the normal cell cluster. We further investigated the CNV SE-score of the reference immune T cells which we used as normal reference. The results showed that the distribution of CNV SE-score in the T cells is similar with the CNV SE-score distribution in cluster2 (Fig. 3E) consistent with the results that cluster2 is the normal cell cluster in the melanoma dataset. In the head-neck cancer data, we found that the four cell clusters had distinct CNV SE-score distributions. The CNV SE-score in cluster4 is significantly higher than other three clusters (Fig. 3I), suggesting that cluster4 might be the normal cell cluster. We further investigated the CNV SE-score of the reference immune T cells which were used as normal reference. The results showed that the distribution of CNV SE-score in the T cells is similar with the CNV SE-score distribution in cluster4 (Fig. 3J) consistent with the results that cluster4 is the normal cell cluster. We also compared the CNV cluster results with the cell labels. Based on the comparison results, we found that almost all the cells in the clusters 1,2,3 are tumor cells, and most of the normal cells were included in the cluster4. In summary, the classification results showed that tumor cells could be predicted by CNV consensus clustering analysis with an accuracy of 0.95 in melanoma data and 0.98 in head-neck cancer data.

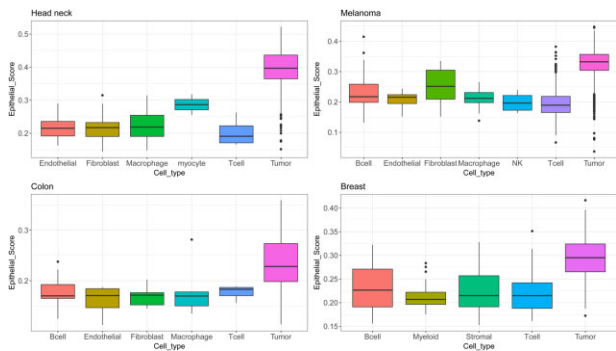


Fig. 4. Distributions of epithelial scores for each cell type in melanoma, head-neck, colon and breast cancer scRNA-seq dataset

To identify cells with active epithelial gene sets in single-cell RNA-seq data, AUCell was used to perform the analysis to quantify the Epithelial Score. We calculated the Epithelial Score of each cell in the melanoma, head neck, colon and breast cancer single-cell datasets. The results showed that tumor cells had a significantly higher Epithelial Score than other normal cells suggesting that the Epithelial Score could be widely applied in different cancer types (Fig. 4). After the Epithelial Score of all cells was calculated, we used the immune T cells as the normal cell reference to calculate the differences (*P*-value) between each cell and the normal distribution. If the Epithelial Score of the cell was significantly higher than the normal distribution, we classified the cell as a tumor cell. Based on this analysis, we found that 0.904 and 0.978 tumor cells in melanoma and head neck data, respectively, could be identified as the tumor cells. To further improve the accuracy for tumor cell classification, we integrated the CNV clustering and Epithelial Score results. The cells which had both copy number variation and significantly higher Epithelial Scores were determined to be malignant cells. The accuracy of the integrated method is 0.96 and 0.99 for tumor cell classification in melanoma and head-neck datasets, respectively. Subsequently all other cells were classified by scDetect. If the cell had both an unknown cell type and copy number variation, we classified the cell as a malignant cell, and the other cells were classified as immune and stromal cell types.

3.4 Performance comparison between scDetect-cancer and other methods in cancer scRNA-seq datasets

Although multiple classification methods had been developed for single-cell type classification, no method has yet to be developed specifically for the single-cell cancer data. In the scDetect-cancer method, we incorporated the copy number variation and epithelial level information to assist in identifying tumor cells. We performed a comparative analysis with CHETHA, SingleR and scPred using the normal cells as the reference in melanoma and head neck data. To make a fair comparison between scDetect-Cancer and other method, we further used CHETHA, scPred and SingleR to replace scDetect in the step of classifying the non-tumour cell types; for identification of tumor cells we used epithelial score and CNV clustering. Here, we called them ‘CHETHA-Cancer’, ‘scPred-Cancer’, ‘SingleR-Cancer’. In the melanoma data, the total cell type prediction accuracy of scDetect-cancer is 0.96, and the prediction precision of tumor cells is 0.98 (Fig. 5A). The prediction accuracy of CHETHA method is 0.75 (Fig. 5C, Supplementary Fig. S2). When we used CHETHA-Cancer for the prediction, the accuracy is 0.95, which is much higher than CHETHA (0.75) but little lower than scDetect-Cancer (0.98). The prediction accuracy of scPred method is 0.90 (Fig. 5C). When we used scPred-Cancer for the prediction, the accuracy is 0.94, which is higher than scPred (0.90) but lower than scDetect-Cancer (0.98). Because the SingleR does not have a rejection option, the tumor cells were seriously misclassified with the Fibroblast cells. Consequently, the accuracy of SingleR is 0.57 in the melanoma dataset (Fig. 5C). When we used SingleR-Cancer for the prediction, the accuracy is 0.93, which is much higher than SingleR (0.57) but lower than scDetect-Cancer (0.98). In the head neck cancer data, the total cell type prediction accuracy of

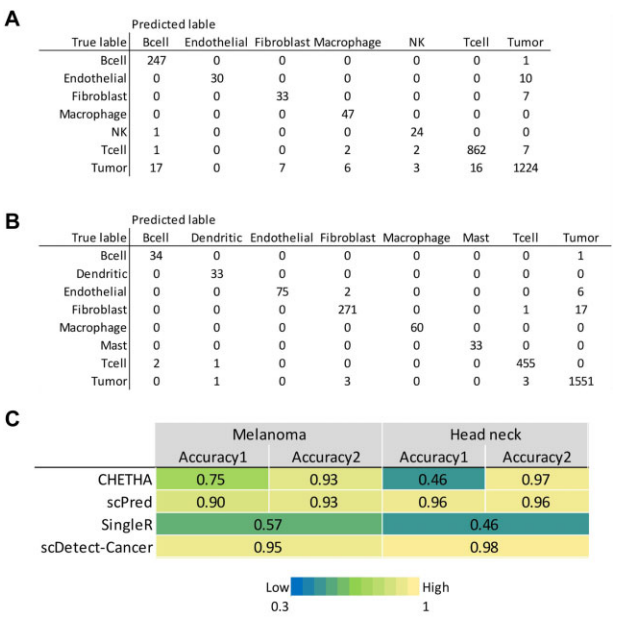


Fig. 5. The cell type classification results of by the scDetect algorithm in melanoma and head-neck cancer datasets. The confusion matrix of cell type classification by the scDetect in (A) melanoma and (B) head-neck datasets. (C) The prediction accuracy comparison between scDetect and other methods. Accuracy1: Unassigned cells were not included in the accuracy calculation; Accuracy2: Unassigned cells were classified as tumor cells in the accuracy calculation

scDetect-cancer is 0.98, and the prediction precision of tumor cells is 0.98 (Fig. 5B). The prediction accuracy of CHETHA method is 0.46 (Fig. 5C). When we used CHETHA-Cancer for the prediction, the accuracy is 0.96. The prediction accuracy of scPred method is 0.96. When we used scPred-Cancer for the prediction, the accuracy is 0.97. Because the SingleR does not have a rejection option, the tumor cells were seriously misclassified as Fibroblast and endothelial cells. The accuracy of SingleR is 0.57 in the head-neck dataset (Fig. 5C). When we used SingleR-Cancer for the prediction, the accuracy is 0.95, which is much higher than SingleR (0.57) but lower than scDetect-Cancer (0.97). In summary, first the study showed that CHETHA-Cancer, scPred-Cancer and SingleR-Cancer performed better than CHETHA, scPred and SingleR in single-cell cancer data cell type classification, which suggested epithelial score and CNV clustering contributed the accuracy in tumor cell classification. Second, scDetect-cancer performed better than CHETHA-Cancer, scPred-Cancer and SingleR-Cancer in cell type classification, which suggested scDetect also contributed the accuracy of scDetect-Cancer in tumor cell classification.

4 Discussion

Single-cell RNA sequencing (scRNA-seq) has provided the ability to analyze the transcriptomic profile of individual cells, leading to the identification of novel cell types and the characterization of heterogeneous cell populations. The demand for a robust method to quantify cell identity continues to grow as technologies to generate scRNA-seq data proliferate and become more accessible. Supervised machine learning methods have been used in previous studies to classify various cell types, such as retinal bipolar cells (Shekhar *et al.*, 2016) and embryonic stem cells (Lin *et al.*, 2017). Here, we introduced scDetect, a gene-rank based weighted majority vote machine learning approach to estimate cell type. In addition to using correlations, the classification decision is also based on a confidence scores determined by the gene rank-based machine learning algorithm. This underlies the highly selective nature of scDetect, facilitating the ability to classify cells as specifically as the input and reference data allow, but without greedy over-classification, as controlled through the confidence score threshold. One consequence is the assignment to unknown cell types for input cells not present in the reference data. The assignment to

unknown cell type is essential to prevent over-classification and acts as an automated flag to more closely inspect such cells. To improve prediction accuracy in the cancer single-cell RNA-seq data, we incorporated cell copy number information and epithelial level information in the scDetect-cancer classification framework for cell type classification. The scDetect-cancer shows high accuracy in a variety of relevant biological and clinical scenarios, including predicting pathological cells from cancer. Collectively, our results show that scDetect/scDetect-cancer is able to accurately classify individual cells from a sample independent of those used to train the prediction model. We have shown that scDetect/scDetect-cancer quantifies cell identity in a manner that is robust across different scRNA-seq platforms. scDetect's confidence threshold can be adjusted to the needs of the dataset at hand, making it a flexible tool for research.

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